

Hydro 1000

Installation and operating instructions



English (GB) Installation and operating instructions

Original installation and operating instructions.

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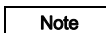
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**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. General information**1.1 Symbols used in this document****Warning**

If these safety instructions are not observed, it may result in personal injury.

**Note**

Notes or instructions that make the job easier and ensure safe operation.

1.2 Applications

Grundfos booster sets Hydro 1000 are designed for the transfer and pressure boosting of clean water in waterworks, blocks of flats, hotels, industry, hospitals, schools, etc.

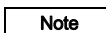
1.3 Operating conditions

Liquid temperature:	Maximum 50 °C.
Ambient temperature:	0-40 °C.
Operating pressure:	Maximum 10 bar.
Minimum and maximum inlet pressures:	See installation and operating instructions for CR pumps.

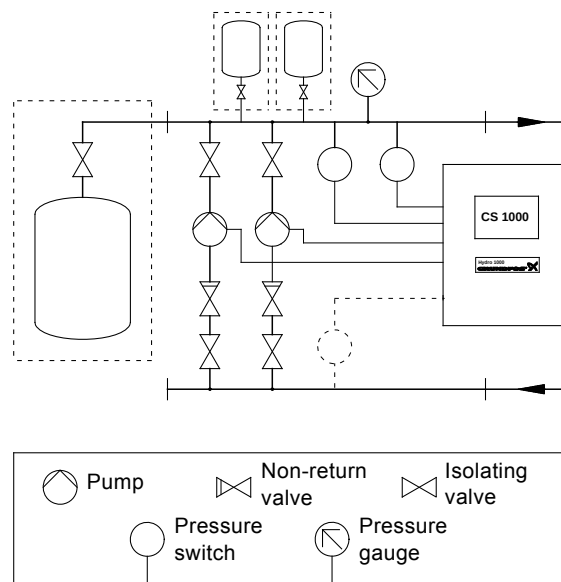
Liquid temperature and operating pressure may be limited by the diaphragm tank.

2. Product description**2.1 Grundfos Hydro 1000**

The booster set consists of 2 to 4 pumps mounted on a common base frame, isolating valves, non-return valves, suction and discharge manifolds with pressure switches and a control box complete with the controller CS 1000.

**Note**

A diaphragm tank must be included in the installation. Diaphragm tanks can be connected directly to the discharge manifold.

Grundfos booster set Hydro 1000**Fig. 1**

2.2 Grundfos Control 1000

The Grundfos Control 1000 controls a number of mains-operated pumps.

The Control 1000 always includes the controller CS 1000 and offers the following functions:

- Automatic cascade control of pumps
- Manual operation
- Pump and system monitoring functions:
 - maximum pressure
 - pre-pressure/level
 - motor protection.
- Display and indication functions:
 - green indicator lights for operating indications and red indicator lights for fault indications
 - potential-free changeover contact for fault indications.

CS 1000 front cover

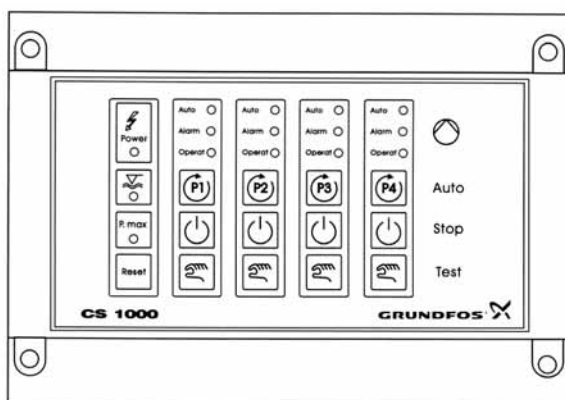


Fig. 2

2.3 Monitoring and indication functions

The following functions are available:

- Red indicator lights (LED) for indication of fault conditions:
 - maximum pressure (if a pressure switch is installed)
 - dry-running protection (if a pressure switch is installed)
 - motor protection.
- Green indicator lights (LED) for indication of operating conditions:
 - power on
 - automatic operation
 - pump in operation.
- Potential-free changeover contact for fault indications.

2.4 Example of booster set Hydro 1000

Grundfos Hydro 1000 CS

Four identical full-size mains-operated pumps and a diaphragm tank.

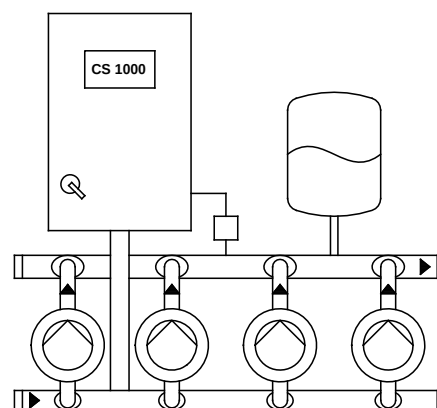


Fig. 3

One pump in operation

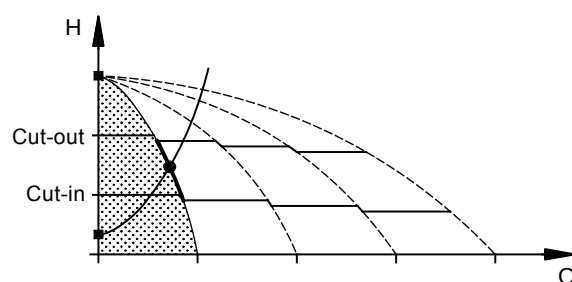


Fig. 4

Three pumps in operation

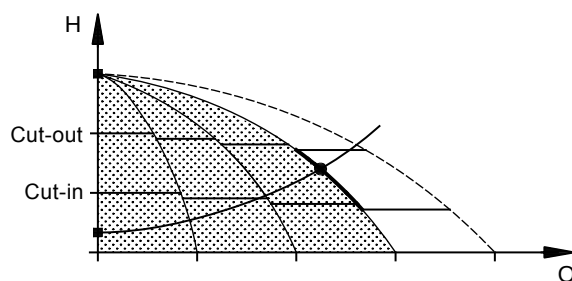


Fig. 5

Grundfos Hydro 1000 CS maintains an almost constant pressure by cutting in/out the required number of pumps.

The CS 1000 alters the starting sequence of the pumps after each pump stop, thus giving all pumps equal operating time.

Pump changeover is automatic and depends on load and fault.

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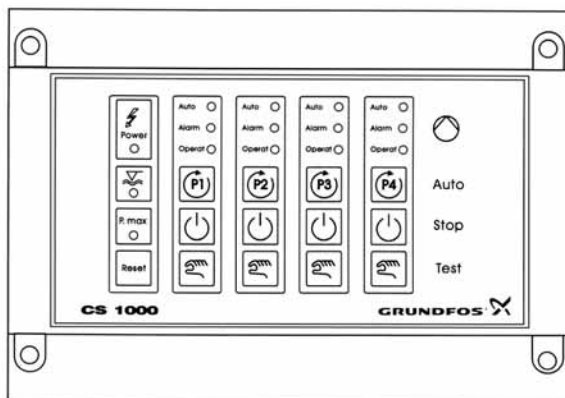
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3. Operation

3.1 CS 1000 front cover

The CS 1000 front cover shows all possible operating conditions.

CS 1000 front cover



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Fig. 6

Button/ indicator light	Function
	Power indication. The green indicator light is on when the electricity supply to the CS 1000 is connected.
	Dry-running alarm. The red indicator light is on when the dry-running alarm is activated. The light goes out when the alarm is reset. If automatic resetting is chosen, the indicator light goes out automatically when normal operating conditions are achieved.
	Maximum pressure alarm. The red indicator light is on when maximum pressure is activated. The light goes out when the alarm is reset. If automatic resetting is chosen, the indicator light goes out automatically when normal operating conditions are achieved.
	Reset button. Resets the dry-running alarm and the maximum pressure alarm if automatic resetting has not been chosen.
Auto ☐	Automatic operation. The green indicator light is on when the pump is released and ready for operation.
Alarm ☐	Pump alarm. The red indicator light is on when the thermal relay of the pump is activated. The pump is out of operation. The alarm can be reset on the thermal relay.
Operat ☐	Pump in operation. The green indicator light is on when the pump is operating.
	Auto button. When the auto button is pressed, the pump will be released for operation and the indicator light for automatic operation will be lit.
	Stop button. When the stop button is pressed, the pump will be taken out of operation and the indicator light for automatic operation goes out.
	Test button. The pump is operating as long as the test button is pressed. The maximum pressure and the dry-running protection are not activated as long as the button is pressed.

4. Installation

4.1 Location

The booster set must be placed in a well-ventilated room to ensure adequate cooling of the pump motors. It must be placed with a 1 metre clearance in front and on the two sides.

Use a forklift truck to move the booster set.

4.2 Foundation

The booster set does not require a special foundation, but it must be positioned on a plane and solid surface, e.g. a concrete floor.

4.3 Pipework

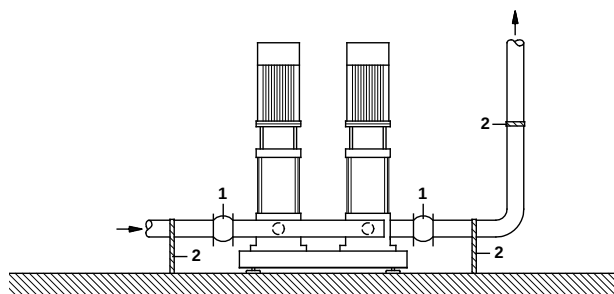
Arrows on the pump bases show the flow direction of the water through the pumps.

The pipes connected to the booster set must be of adequate size. To avoid resonance, expansion joints must be fitted in the suction and discharge pipes. See fig. 7.

The pipes are connected to the manifolds of the booster set. Either end can be used. Apply sealing compound to the unused end of the manifold and fit a screw cap. For manifolds with flanges, a blind flange with gasket is fitted.

The booster set must be tightened up before start-up.

The pipes must be fastened to parts of the building to ensure that they cannot move or be twisted.



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Fig. 7

1. Expansion joint.
2. Pipe hanger.

Expansion joints, pipe hangers and vibration dampers shown in fig. 7 are not included in the standard booster set.

4.4 Dry-running protection

It is possible to install a pressure switch on the suction side of the booster set to monitor the inlet pressure and to prevent the booster set from running dry (a level switch can be used if the suction side is connected to an open tank).

The booster set must not be started until a dry-running protection has been installed. When the inlet pressure/level drops below the set value, the pumps are cut out. When the correct pressure/level value is restored, the pumps can be reset to operation automatically or manually.

4.5 Priming

Do not start the pumps until they have been filled with water. See installation and operating instructions for CR pumps.

4.6 Electrical connection

The electrical connection must be carried out by an authorized electrician in accordance with local regulations.

The wiring diagram and motor labels show required values for the electricity supply.

No additional motor protection is required as the control box incorporates motor protection.



Warning

Before making any connections in pumps, terminal boxes or control box, the electricity supply must be switched off.

5. Settings

5.1 Timers

The three timers incorporated in the controller CS 1000 can be used to optimize the operation of the booster set. The timers are to be adjusted individually.

Start-up delay TR 1 (0-5 secs.)

Defined as: The minimum time between starts of individual pumps.

- Prevents:
- simultaneous start of all pumps in case of:
 - momentary power failure
 - peak of water demand
 - automatic resetting of dry-running
 - negative pressure in suction manifold
 - overload of power mains
 - water hammer.

Stop delay TR 2 (0-5 secs.)

Defined as: The minimum time between stops of individual pumps.

- Prevents:
- simultaneous stop of all pumps in case of:
 - rapid fall of water demand
 - water hammer.

Reduces: number of starts and stops per hour.

After-run delay TR 3 (0-5 secs.)

Defined as: The time that pumps are kept in operation after the cut-out pressure is reached.

Keeps: pumps in operation when the cut-out pressure is reached.

Reduces: number of starts and stops per hour.

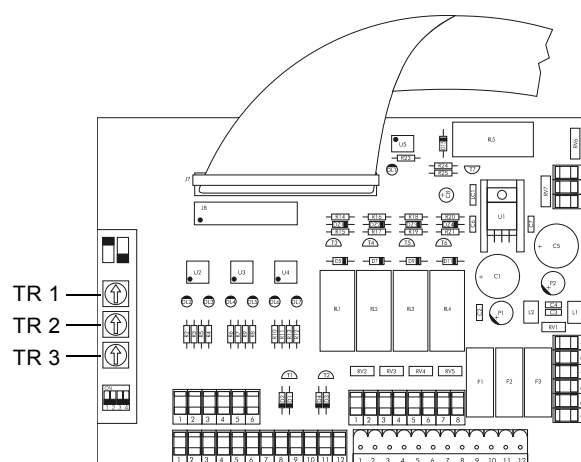


Fig. 8

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5.2 Operation selection

All operating possibilities for the pumps in your booster set are listed below:

To select operation, proceed as follows:

- Switch off the electricity supply to the Control 1000.
- Open the control box and set the DIP switches according to the desired selection.
- Close the control box, switch on the electricity supply and press buttons as indicated in column "Front panel selection".



Warning
Before changing DIP switch setting, switch off the electricity supply.



Fig. 9

Number of pumps installed	Front panel selection	DIP switch setting	Pumps in operation	Pumps to be alternated	Pilot pump
4 pumps	 P1 P2 P3 P4		P1 P2 P3 P4	P1 P2 P3 P4	
	 — — — —		P1 P2 P3 P4	— — — —	
	 — — — —		P1 P2 P3 P4	— P2 P3 P4	P1
3 pumps	 P1 P2 P3		P1 P2 P3	P1 P2 P3	
	 P4		P1 P2 P3	— — —	
	 P4		P1 P2 P3	— P2 P3	P1
2 pumps	 P1 P2		P1 P2	P1 P2	
	 P3 P4		P1 P2	— —	
	 P3 P4				

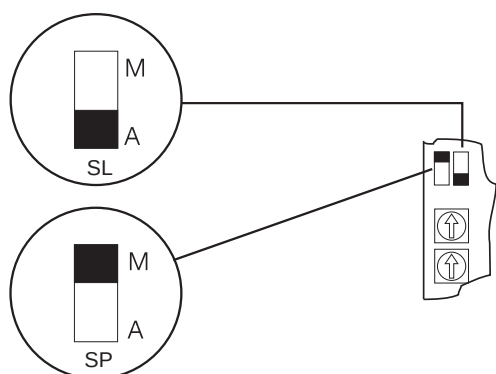
- A pilot pump is the pump
- that tarts first/stops last
 - that is generally smaller than the other pumps (smaller flow - higher head)
 - that is not to be alternated
 - that must be installed in position P1.

Note

5.3 Alarm resetting

There are two possible resetting modes which can be selected by means of each of the switches SL and SP.

- SL: Dry running
- SP: High pressure.



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Fig. 10

- M (Manual). When acceptable conditions have been restored, the pumps are started by pressing the reset button on the front cover.
- A (Automatic). When the pressure/level has reached an acceptable value, the pumps start automatically.

5.4 Pressure setting

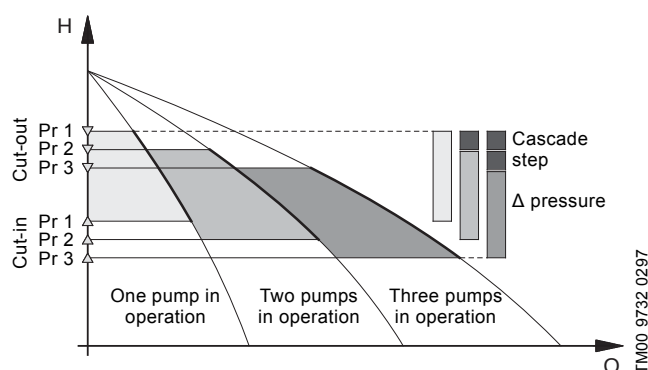
The pumps are cut in and out according to the pressures set on the pressure switches.

A common differential (Δ) pressure (cut-out pressure - cut-in pressure = Δ pressure) must be set on the pressure switches. The cut-out pressures differ 0.3 to 0.5 bar respectively (cascade step).

Example:

- 3 pumps.
- Δ pressure = 1.5 bar.
- Cascade step = 0.3 bar.
- Pr 1, 2, 3 = Pressure switches 1, 2, 3.
- P 1, 2, 3 = Pumps 1, 2, 3.

Pressure switch and pump numbers	Cut-in pressure [bar]	Cut-out pressure [bar]	Δ pressure [bar]
Pr 1 / P 1	2.6	4.1	1.5
Pr 2 / P 2	2.3	3.8	1.5
Pr 3 / P 3	2.0	3.5	1.5



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Fig. 11

The settings of the pressure switches Pr 2 and Pr 3 are in a decreasing sequence as shown in the table and fig. 11.

When water is consumed, this is first tapped from the diaphragm tank. Then the pressure drops to the first cut-in pressure, and the first pump is cut in. If the consumption is still rising, more pumps will be cut in until the performance of the pumps in operation corresponds to the requirements.

When the water consumption falls, the discharge pressure rises to the cut-out pressure and the CS 1000 cuts out one pump. If the consumption is still falling, more pumps will be cut out.

Note

Each pressure switch is related to one pump only and not necessarily to the same pump when pump alternation is selected on the DIP switch, see section 5.2 Operation selection.

Pressure switch Pr 1 must be set to the highest cut-in and cut-out values.

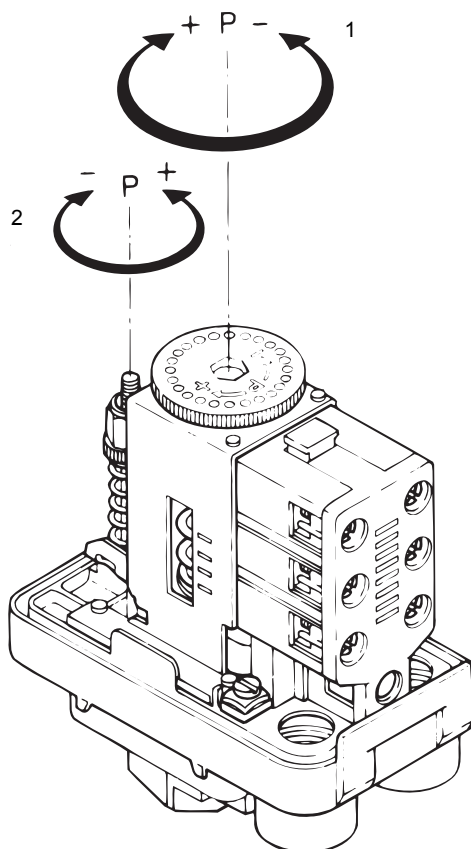
5.5 Pressure switch

On each pressure switch, the following settings must be made:

- cut-out pressure
- Δ pressure (differential pressure).

Note

The cut-in pressure is set as the cut-out pressure minus Δ pressure.



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Fig. 12

1. Cut-out pressure:
+ to increase
- to decrease.
2. Δ pressure (differential pressure):
+ to increase
- to decrease.

5.6 Tank precharge pressure

The precharge pressure is equal to the minimum cut-in pressure.

6. Startup

1. Connect water and electricity supplies.
2. Check that the precharge pressure in the diaphragm tank is equal to the minimum cut-in pressure.
3. Close the discharge valves of the pumps and prime the booster set and the suction pipe.
4. Adjust the pressure switches according to the desired discharge pressure.
5. Check that the DIP switch setting corresponds to the number of pumps installed.
6. Switch on the electricity supply.
7. Start pump 1 by pressing the button on the front cover.



8. Pump 1 will now start.
9. Check the direction of rotation of the pump. In case of incorrect rotation, interchange two phases of the electricity supply.
10. Vent the pump while slowly opening its discharge valve.
11. Repeat points 7 to 10 for all pumps.
12. Release the pumps for operation by pressing the buttons on the front cover.



Hydro 1000 is now ready for operation.

7. Maintenance

7.1 Maintenance of booster set

See installation and operating instructions for CR pumps.

7.2 Maintenance of Control 1000

The Control 1000 is maintenance-free.

7.3 Checking the precharge pressure

To ensure reliable and correct operation, including the frequency of starts and stops, the precharge pressure of the diaphragm tank must be checked regularly (at least twice a year).

8. Fault finding



Warning

Before making any connections in pumps, terminal boxes or control box, the power supply must be switched off.

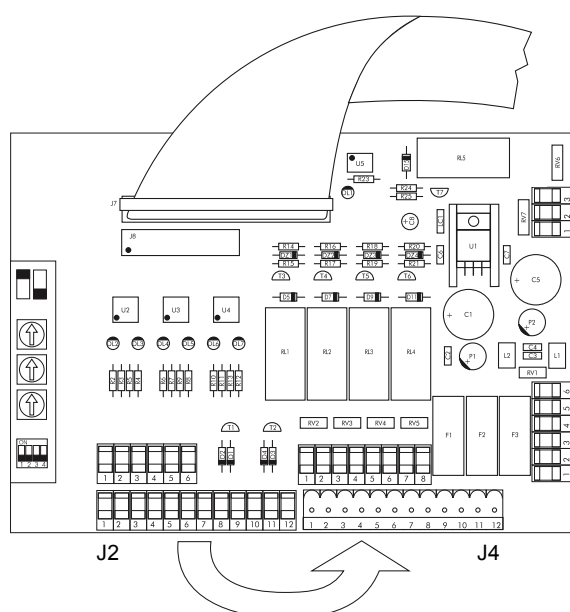
Fault	Cause	Remedy
1. Motor does not run when started.	a) Electricity supply disconnected.	Connect the electricity supply.
	b) Automatic circuit breakers cut out.	Correct the fault and cut in the automatic circuit breakers.
	c) Fuse in the CS 1000 defective.	Replace the fuse.
	d) Motor defective.	Repair/replace the motor.
2. Motor starts, but stops immediately afterwards.	a) Incorrect setting of pressure switch.	Increase the cut-out pressure and/or differential pressure.
	b) Wrong diaphragm tank precharge pressure.	Check the diaphragm tank precharge pressure.
3. Frequent starts and stops.	a) Incorrect setting of time delay TR 2.	Set the time delay TR 2 accordingly.
	b) Incorrect setting of pressure switch.	Increase the cut-out pressure and/or differential pressure.
	c) Diaphragm tank defective.	Repair the diaphragm tank.
4. Pumps are running but deliver no water.	a) Suction pipe/pumps blocked by impurities.	Clean the suction pipe/pumps.
	b) Non-return valve blocked in closed position.	Clean the non-return valve.
	c) Suction pipe leaky.	Repair the suction pipe.
	d) Air in suction pipe/pumps.	Vent the pumps. Check the suction pipe for leakages.
	e) Motors running with the wrong direction of rotation.	Change the direction of rotation.
5. Leakage from the shaft seal.	a) Shaft seal defective.	Replace the shaft seal.
	b) Height adjustment of the pump shaft inaccurate.	Readjust the shaft height. See installation and operating instructions for CR pumps.
6. Noise.	a) The pumps are cavitating.	Clean the suction pipe/pumps and possibly the suction strainer.
	b) The pumps do not rotate freely (frictional resistance) due to inaccurate height adjustment of the pump shaft.	Readjust the shaft height. See installation and operating instructions for CR pumps.

8.1 Emergency operation

In case of damage to electronic components belonging to the printed circuit board, the system can be operated by moving the multiplug from J2 to J4, see fig. 13.

Pumps will still operate automatically by pressure switches but without LED indications and the alternation function.

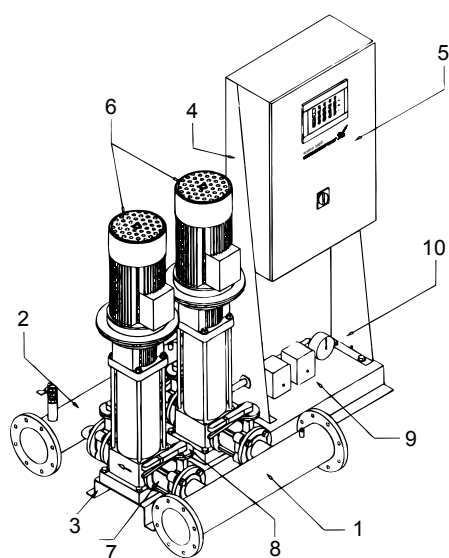
Maximum pressure and dry-running protection are not active when the booster set is in emergency operation.



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Fig. 13

9. Parts list



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Fig. 14

Pos.	Description	Number
1	Suction manifold	1
2	Discharge manifold	1
3	Base frame	1
4	Stand	1
5	Control box	1
6	CR pump	2-4
7	Non-return valve	1 per pump
8	Isolating valve	2 per pump
9	Pressure switch	1 per pump
10	Pressure gauge	1

10. Technical data

10.1 Sound pressure level

For sound pressure level, see installation and operating instructions for CR pumps.

The sound pressure level for a number of pumps can be calculated as follows:

$$L_{\max} = L_{\text{pump}} + (n - 1) \times 3.$$

$L_{\max}$ = Maximum sound pressure level.

L_{pump} = Sound pressure level for one pump.

n = Number of pumps.

10.2 Hydraulic data

Minimum pre-pressure:

The minimum pre-pressure "H" in metres head required to avoid cavitation in the pumps is calculated as follows:

$$H = p_b \times 10.2 - \text{NPSH} - H_f - H_v - H_s$$

p_b = Barometric pressure in bar.

Barometric pressure can be set to 1 bar.

NPSH = **N**et **P**ositive **S**uction **H**ead i metres head (to be read from the NPSH curve at the highest flow).
See installation and operating instructions for CR pumps.

H_f = Friction loss in suction pipe in metres head.

H_v = Vapour pressure in metres head, t_m = liquid temperature.
See installation and operating instructions for CR pumps.

H_s = Safety margin of minimum 0.5 metres head.

10.3 Electrical data

Supply Voltage: 3 x 400 V + 6 %/- 10 %, 50 Hz, PE.

Terminal designation or number	Function CS 1000	Technical data
1	Input for thermal relay pump 1	
2	Input for thermal relay pump 2	
3	Input for thermal relay pump 3	
J1 4	Input for thermal relay pump 4	
5		
6	Common terminal for multiplug J1	
1-2	Input for dry-running protection	
3-4	Input for maximum pressure protection	
J2 5-6	Input for pressure switch 1	
7-8	Input for pressure switch 2	
9-10	Input for pressure switch 3	
11-12	Input for pressure switch 4	
1-2	Output for contactor K1	AC 24 V
J3 3-4	Output for contactor K2	AC 24 V
5-6	Output for contactor K3	AC 24 V
7-8	Output for contactor K4	AC 24 V
1-2	Electricity supply for CS 1000	AC 24 V
J5 3-4	Electricity supply for CS 1000	AC 18 V
5-6	Electricity supply for CS 1000	AC 8 V
		Potential-free change-over contacts.
J6 1-3	Fault relay	Maximum contact load: 250 V / 3.0 A, AC 1

11. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

Declaration of conformity

EC/EU declaration of conformity

We, Grundfos, declare under our sole responsibility that the product Hydro 1000, to which the declaration below relates, is in conformity with the Council Directives listed below on the approximation of the laws of the EC/EU member states.

Note: There are two sets of Council Directives and standards listed below. One set applies until and including 19th April 2016. The other set applies from 20th April 2016 and onwards.

These Directives and standards apply until and including 19th April 2016:

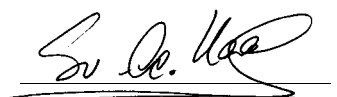
- Machinery Directive (2006/42/EC).
Standard used: EN 809 + A1/AC: 2010.
- EMC Directive (2004/108/EC).
Standards used: EN 61000-6-2: 2005, EN 61000-6-3: 2007.

These Directives and standards apply from 20th April 2016 and onwards:

- Machinery Directive (2006/42/EC).
Standard used: EN 809 + A1/AC: 2010.
- EMC Directive (2014/30/EU).
Standards used: EN 61000-6-2: 2005, EN 61000-6-3: 2007.

This EC/EU declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication number 96413144 1115).

Bjerringbro, 29 October 2015



Svend Aage Kaae
Director
Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro, Denmark

Person authorised to compile the technical file and
empowered to sign the EC/EU declaration of conformity.

Argentina

Bombas GRUNDFOS de Argentina S.A.
Ruta Panamericana km. 37.500 Centro
Industrial Garin
1619 Garin Pcia. de B.A.
Phone: +54-3327 414 444
Telefax: +54-3327 45 3190

Australia

GRUNDFOS Pumps Pty. Ltd.
P.O. Box 2040
Regency Park
South Australia 5942
Phone: +61-8-8461-4611
Telefax: +61-8-8340 0155

Austria

GRUNDFOS Pumpen Vertrieb Ges.m.b.H.
Grundfosstraße 2
A-5082 Grödig/Salzburg
Tel.: +43-6246-883-0
Telefax: +43-6246-883-30

Belgium

N.V. GRUNDFOS Bellux S.A.
Boomsesteenweg 81-83
B-2630 Aartselaar
Tél.: +32-3-870 7300
Télécopie: +32-3-870 7301

Belarus

Представительство ГРУНДФОС в
Минске
220125, Минск
ул. Шафарьянская, 11, оф. 56, 5Ц
«Порт»
Тел.: +7 (375 17) 286 39 72/73
Факс: +7 (375 17) 286 39 71
E-mail: minsk@grundfos.com

Bosna and Herzegovina

GRUNDFOS Sarajevo
Zmaja od Bosne 7-7A,
BH-71000 Sarajevo
Phone: +387 33 592 480
Telefax: +387 33 590 465
www.ba.grundfos.com
e-mail: grundfos@bih.net.ba

Brazil

BOMBAS GRUNDFOS DO BRASIL
Av. Humberto de Alencar Castelo Branco,
630
CEP 09850 - 300
São Bernardo do Campo - SP
Phone: +55-11 4393 5533
Telefax: +55-11 4343 5015

Bulgaria

Grundfos Bulgaria EOOD
Slatina District
Iztochna Tangenta street no. 100
BG - 1592 Sofia
Tel. +359 2 49 22 200
Fax. +359 2 49 22 201
email: bulgaria@grundfos.bg

Canada

GRUNDFOS Canada Inc.
2941 Brighton Road
Oakville, Ontario
L6H 6C9
Phone: +1-905 829 9533
Telefax: +1-905 829 9512

China

GRUNDFOS Pumps (Shanghai) Co. Ltd.
10F The Hub, No. 33 Suhong Road
Minhang District
Shanghai 201106
PRC
Phone: +86 21 612 252 22
Telefax: +86 21 612 253 33

Croatia

GRUNDFOS CROATIA d.o.o.
Buzinski prilaz 38, Buzin
HR-10010 Zagreb
Phone: +385 1 6595 400
Telefax: +385 1 6595 499
www.hr.grundfos.com

Czech Republic

GRUNDFOS s.r.o.
Čajkovského 21
779 00 Olomouc
Phone: +420-585-716 111
Telefax: +420-585-716 299

Denmark

GRUNDFOS DK A/S
Martin Bachs Vej 3
DK-8850 Bjerringbro
Tlf.: +45-87 50 50 50
Telefax: +45-87 50 51 51
E-mail: info_GDK@grundfos.com
www.grundfos.com/DK

Estonia

GRUNDFOS Pumps Eesti OÜ
Peterburi tee 92G
11415 Tallinn
Tel: + 372 606 1690
Fax: + 372 606 1691

Finland

OY GRUNDFOS Pumput AB
Trukkikuja 1
FI-01360 Vantaa
Phone: +358-(0) 207 889 500
Telefax: +358-(0) 207 889 550

France

Pompes GRUNDFOS Distribution S.A.
Parc d'Activités de Chesnes
57, rue de Malacombe
F-38290 St. Quentin Fallavier (Lyon)
Tél.: +33-4 74 82 15 15
Télécopie: +33-4 74 94 10 51

Germany

GRUNDFOS GMBH
Schlüterstr. 33
40699 Erkrath
Tel.: +49-(0) 211 929 69-0
Telefax: +49-(0) 211 929 69-3799
e-mail: infoservice@grundfos.de
Service in Deutschland:
e-mail: kundendienst@grundfos.de

Greece

GRUNDFOS Hellas A.E.B.E.
20th km. Athinon-Markopoulou Av.
P.O. Box 71
GR-19002 Peania
Phone: +0030-210-66 83 400
Telefax: +0030-210-66 46 273

Hong Kong

GRUNDFOS Pumps (Hong Kong) Ltd.
Unit 1, Ground floor
Siu Wai Industrial Centre
29-33 Wing Hong Street &
68 King Lam Street, Cheung Sha Wan
Kowloon
Phone: +852-27861706 / 27861741
Telefax: +852-27858664

Hungary

GRUNDFOS Hungária Kft.
Park u. 8
H-2045 Törökbálint,
Phone: +36-23 511 110
Telefax: +36-23 511 111

India

GRUNDFOS Pumps India Private Limited
118 Old Mahabalipuram Road
Thoraiakkam
Chennai 600 096
Phone: +91-44 2496 6800

Indonesia

PT. GRUNDFOS POMPA
Graha Intirub Lt. 2 & 3
Jln. Cililitan Besar No.454. Makasar,
Jakarta Timur
ID-Jakarta 13650
Phone: +62 21-469-51900
Telefax: +62 21-460 6910 / 460 6901

Ireland

GRUNDFOS (Ireland) Ltd.
Unit A, Merrywell Business Park
Ballymount Road Lower
Dublin 12
Phone: +353-1-4089 800
Telefax: +353-1-4089 830

Italy

GRUNDFOS Pompe Italia S.r.l.
Via Gran Sasso 4
I-20060 Truccazzano (Milano)
Tel.: +39-02-95838112
Telefax: +39-02-95309290 / 95838461

Japan

GRUNDFOS Pumps K.K.
Gotanda Metalon Bldg., 5F,
5-21-15, Higashi-gotanda
Shiagawa-ku, Tokyo
141-0022 Japan
Phone: +81 35 448 1391
Telefax: +81 35 448 9619

Korea

GRUNDFOS Pumps Korea Ltd.
6th Floor, Aju Building 679-5
Yeoksam-dong, Kangnam-ku, 135-916
Seoul, Korea
Phone: +82-2-5317 600
Telefax: +82-2-5633 725

Latvia

SIA GRUNDFOS Pumps Latvia
Deglava biznesa centrs
Augusta Deglava ielā 60, LV-1035, Rīga,
Tālr.: + 371 714 9640, 7 149 641
Fakss: + 371 914 9646

Lithuania

GRUNDFOS Pumps UAB
Smolensko g. 6
LT-03201 Vilnius
Tel: + 370 52 395 430
Fax: + 370 52 395 431

Malaysia

GRUNDFOS Pumps Sdn. Bhd.
7 Jalan Peguam U1/25
Glenmarie Industrial Park
40150 Shah Alam
Selangor
Phone: +60-3-5569 2922
Telefax: +60-3-5569 2866

Mexico

Bombas GRUNDFOS de México S.A. de
C.V.
Boulevard TLC No. 15
Parque Industrial Stiva Aeropuerto
Apodaca, N.L. 66600
Phone: +52-81-8144 4000
Telefax: +52-81-8144 4010

Netherlands

GRUNDFOS Netherlands
Veluwezoom 35
1326 AE Almere
Postbus 22015
1302 CA ALMERE
Tel.: +31-88-478 6336
Telefax: +31-88-478 6332
E-mail: info_gnl@grundfos.com

New Zealand

GRUNDFOS Pumps NZ Ltd.
17 Beatrice Tinsley Crescent
North Harbour Industrial Estate
Albany, Auckland
Phone: +64-9-415 3240
Telefax: +64-9-415 3250

Norway

GRUNDFOS Pumper A/S
Strømsveien 344
Postboks 235, Leirdal
N-1011 Oslo
Tlf.: +47-22 90 47 00
Telefax: +47-22 32 21 50

Poland

GRUNDFOS Pompy Sp. z o.o.
ul. Klonowa 23
Baranowo k. Poznania
PL-62-081 Przeźmierowo
Tel: (+48-61) 650 13 00
Fax: (+48-61) 650 13 50

Portugal

Bombas GRUNDFOS Portugal, S.A.
Rua Calvet de Magalhães, 241
Apartado 1079
P-2770-153 Paço de Arcos
Tel.: +351-21-440 76 00
Telefax: +351-21-440 76 90

Romania

GRUNDFOS Pompe România SRL
Bd. Biruintei, nr 103
Pantelimon county Ilfov
Phone: +40 21 200 4100
Telefax: +40 21 200 4101
E-mail: romania@grundfos.ro

Russia

ООО Грундфос Россия
109544, г. Москва, ул. Школьная, 39-41,
стр. 1
Тел. (+7) 495 564-88-00 (495) 737-30-00
Факс (+7) 495 564 88 11
E-mail grundfos.moscow@grundfos.com

Serbia

Grundfos Srbija d.o.o.
Omladinskih brigada 90b
11070 Novi Beograd
Phone: +381 11 2258 740
Telefax: +381 11 2281 769
www.rs.grundfos.com

Singapore

GRUNDFOS (Singapore) Pte. Ltd.
25 Jalan Tukang
Singapore 619264
Phone: +65-6681 9688
Telefax: +65-6681 9689

Slovakia

GRUNDFOS s.r.o.
Prievozská 4D
821 09 BRATISLAVA
Phona: +421 2 5020 1426
sk.grundfos.com

Slovenia

GRUNDFOS d.o.o.
Šlandrova 8b, SI-1231 Ljubljana-Črnuče
Phone: +386 31 718 808
Telefax: +386 (0)1 5680 619
E-mail: slovenia@grundfos.si

South Africa

GRUNDFOS (PTY) LTD
Corner Mountjoy and George Allen Roads
Wilbart Ext. 2
Bedfordview 2008
Phone: (+27) 11 579 4800
Fax: (+27) 11 455 6066
E-mail: lsmart@grundfos.com

Spain

Bombas GRUNDFOS España S.A.
Camino de la Fuentequilla, s/n
E-28110 Algete (Madrid)
Tel.: +34-91-848 8800
Telefax: +34-91-628 0465

Sweden

GRUNDFOS AB
Box 333 (Lunnagårdsgatan 6)
431 24 Mölndal
Tel.: +46 31 332 23 000
Telefax: +46 31 331 94 60

Switzerland

GRUNDFOS Pumpen AG
Bruggacherstrasse 10
CH-8117 Fällanden/ZH
Tel.: +41-44-806 8111
Telefax: +41-44-806 8115

Taiwan

GRUNDFOS Pumps (Taiwan) Ltd.
7 Floor, 219 Min-Chuan Road
Taichung, Taiwan, R.O.C.
Phone: +886-4-2305 8899
Telefax: +886-4-2305 0878

Thailand

GRUNDFOS (Thailand) Ltd.
92 Chaloem Phrakiat Rama 9 Road,
Dokmai, Pravej, Bangkok 10250
Phone: +66-2-725 8999
Telefax: +66-2-725 8998

Turkey

GRUNDFOS POMPA San. ve Tic. Ltd. Sti.
Gebze Organize Sanayi Bölgesi
İhsan dede Caddesi,
2. yol 200, Sokak No. 204
41490 Gebze/ Kocaeli
Phone: +90 - 262-679 7979
Telefax: +90 - 262-679 7905
E-mail: satis@grundfos.com

Ukraine

Бізнес Центр Європа
Столичне шосе, 103
м. Київ, 03131, Україна
Телефон: (+38 044) 237 04 00
Факс: (+38 044) 237 04 01
E-mail: ukraine@grundfos.com

United Arab Emirates

GRUNDFOS Gulf Distribution
P.O. Box 16768
Jebel Ali Free Zone
Dubai
Phone: +971 4 8815 166
Telefax: +971 4 8815 136

United Kingdom

GRUNDFOS Pumps Ltd.
Grovebury Road
Leighton Buzzard/Beds. LU7 4TL
Phone: +44-1525-850000
Telefax: +44-1525-850011

U.S.A.

GRUNDFOS Pumps Corporation
17100 West 118th Terrace
Olathe, Kansas 66061
Phone: +1-913-227-3400
Telefax: +1-913-227-3500

Uzbekistan

Grundfos Tashkent, Uzbekistan The Repre-
sentative Office of Grundfos Kazakhstan in
Uzbekistan
38a, Oybek street, Tashkent
Телефон: (+998) 71 150 3290 / 71 150
3291
Факс: (+998) 71 150 3292

Addresses Revised 29.09.2015

96413144 1115
ECM: 1168508

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